

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>CPU Scheduling Simulator | Preemptive SJF</title>


  <link rel="preconnect" href="https://fonts.googleapis.com">

  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>

  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap"
rel="stylesheet">

  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.4.0/css/all.min.css">


<style>

/* 🌿 Soft Mobile App Palette (Teal, Light Yellow, Cream) */

:root {

  --primary: #38A9AC;    /* Main Teal */

  --secondary: #41ABAE;  /* Mint Teal */

  --accent: #FDF4A9;     /* Soft Pastel Yellow */


  --bg-color: #F4F8F8;   /* Very Soft Cream/Teal Background */

  --text-main: #1B4332;  /* Very Dark Teal for high contrast */

  --text-muted: #527A6C; /* Muted Dark Teal */


  --card-bg: #FFFFFF;    /* Pure white for modern cards */

  --card-border: #E8F0F0;

  --nav-bg: rgba(255, 255, 255, 0.85);

  --input-bg: #F9FBFB;


  /* Deep, soft shadows for the floating app feel */
```

```

--shadow-soft: 0 10px 30px rgba(36, 86, 67, 0.06);

--shadow-hover: 0 20px 40px rgba(56, 169, 172, 0.12);

--shadow-btn: 0 8px 20px rgba(56, 169, 172, 0.25);
}

/* Global Reset & Typography */
* { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Inter', sans-serif; }

body {
  background-color: var(--bg-color);
  color: var(--text-main);
  line-height: 1.7;
  min-height: 100vh;
  overflow-x: hidden;
  padding-bottom: 2rem;
}

/* 🧹 Clean Minimal Navigation */
nav {
  display: flex; justify-content: space-between; align-items: center;
  padding: 1.2rem 5%;
  background: var(--nav-bg);
  backdrop-filter: blur(12px);
  -webkit-backdrop-filter: blur(12px);
  border-bottom: 1px solid var(--card-border);
  position: sticky; top: 0; z-index: 100;
  box-shadow: 0 4px 20px rgba(0,0,0,0.02);
}

.logo {
  font-size: 1.25rem; font-weight: 800;
  color: var(--text-main);
  display: flex; align-items: center; gap: 10px;

```

```

    letter-spacing: -0.3px;
}

.logo i { color: var(--primary); font-size: 1.4rem; }

.nav-links { display: flex; gap: 0.5rem; background: var(--input-bg); padding: 0.4rem; border-
radius: 50px; border: 1px solid var(--card-border); }

.nav-links button {
    background: transparent; border: none; font-size: 0.95rem;
    color: var(--text-muted); cursor: pointer; font-weight: 600;
    padding: 0.6rem 1.4rem; border-radius: 50px;
    transition: all 0.3s ease;
}

.nav-links button:hover { color: var(--primary); background: #E8F4F4; }

.nav-links button.active {
    background: var(--accent);
    color: var(--text-main);
    box-shadow: 0 4px 10px rgba(253, 244, 169, 0.5);
}

/* ✨ Animations */
@keyframes fadeUp {
    from { opacity: 0; transform: translateY(30px); }
    to { opacity: 1; transform: translateY(0); }
}

@keyframes float {
    0%, 100% { transform: translateY(0); }
    50% { transform: translateY(-15px); }
}

@keyframes popIn {
    0% { opacity: 0; transform: scale(0.9); }
    100% { opacity: 1; transform: scale(1); }
}

```

```
}
```

```
/* Page Layout */
```

```
.page { display: none; padding: 4rem 5%; max-width: 1100px; margin: 0 auto; opacity: 0; }
```

```
.page.active { display: block; animation: fadeUp 0.6s cubic-bezier(0.16, 1, 0.3, 1) forwards; }
```

```
h1, h2, h3 { color: var(--text-main); font-weight: 700; letter-spacing: -0.5px; }
```

```
h2 { font-size: 1.8rem; margin-bottom: 2rem; display: flex; align-items: center; gap: 0.75rem; color: var(--primary); }
```

```
h3 { color: var(--text-main); font-size: 1.2rem; }
```

```
p { color: var(--text-muted); margin-bottom: 1.5rem; font-size: 1.05rem; }
```

```
/* 🌟 Modern Rounded Cards */
```

```
.card {
```

```
  background: var(--card-bg);
```

```
  border-radius: 24px; padding: 2.5rem;
```

```
  box-shadow: var(--shadow-soft); margin-bottom: 2rem;
```

```
  border: 1px solid rgba(255,255,255,0.8);
```

```
  transition: all 0.4s ease;
```

```
}
```

```
.card:hover { transform: translateY(-6px); box-shadow: var(--shadow-hover); }
```

```
.card h3 { margin-bottom: 1.5rem; display: flex; align-items: center; gap: 10px; }
```

```
.grid-2 { display: grid; grid-template-columns: repeat(auto-fit, minmax(300px, 1fr)); gap: 2rem; }
```

```
/* 🍯 Pill-Shaped Buttons */
```

```
.btn {
```

```
  background: var(--primary);
```

```
  color: white; border: none; padding: 0.9rem 2rem;
```

```
  border-radius: 50px; font-weight: 600; font-size: 1rem;
```

```
  cursor: pointer; transition: all 0.3s ease;
```

```
  display: inline-flex; align-items: center; justify-content: center; gap: 0.5rem;
```

```

        box-shadow: var(--shadow-btn);
    }

    .btn:hover { background: var(--secondary); transform: scale(1.04) translateY(-2px); box-shadow:
0 10px 20px rgba(56, 169, 172, 0.35); }

    .btn:active { transform: scale(0.98); }

    .btn-outline {

        background: transparent; color: var(--primary);

        border: 2px solid var(--primary); box-shadow: none;
    }

    .btn-outline:hover { background: var(--input-bg); color: var(--primary); box-shadow: 0 4px 12px
rgba(56, 169, 172, 0.1); }

/* Form Inputs */

.input-group { display: flex; flex-direction: column; gap: 0.5rem; margin-bottom: 1.5rem; }

.input-group label { font-weight: 600; font-size: 0.9rem; color: var(--text-main); margin-left:
0.5rem; }

    input {

        padding: 1.1rem 1.2rem; border: 2px solid var(--card-border);

        border-radius: 20px; background: var(--input-bg);

        color: var(--text-main); font-size: 1rem; font-weight: 500;

        transition: all 0.3s ease; outline: none;
    }

    input:focus { border-color: var(--primary); background: #FFFFFF; box-shadow: 0 0 0 4px rgba(56,
169, 172, 0.15); }

/* Clean Tables (General) */

table { width: 100%; border-collapse: separate; border-spacing: 0 8px; margin-top: 1rem; }

th { color: var(--text-muted); font-size: 0.85rem; text-transform: uppercase; letter-spacing: 1px;
font-weight: 700; padding: 0.5rem 1rem; border: none; }

td { background: var(--input-bg); padding: 1.2rem 1rem; border: none; font-weight: 500;
transition: all 0.3s ease; }

td:first-child { border-radius: 16px 0 0 16px; }

```

```

td:last-child { border-radius: 0 16px 16px 0; }

tbody tr:hover td { background: var(--card-border); transform: scale(1.01); }

/* Gantt Chart (General Framework) */

.gantt-container { display: flex; flex-direction: column; margin-top: 2rem; overflow-x: auto;
padding-bottom: 2rem; }

.gantt-chart { display: flex; height: 70px; border-radius: 20px; padding: 6px; background: var(--
input-bg); border: 1px solid var(--card-border); }

.gantt-block {

    display: flex; align-items: center; justify-content: center;

    color: var(--text-main); font-weight: 600; font-size: 1rem;

    position: relative; transition: width 0.6s cubic-bezier(0.2, 0.8, 0.2, 1);

    border-radius: 14px;

    border: 2px solid var(--input-bg);

    overflow: hidden;

    animation: popIn 0.5s ease-out forwards;

}

/* Metrics Grid */

.metrics { display: grid; grid-template-columns: repeat(auto-fit, minmax(220px, 1fr)); gap:
1.5rem; margin-bottom: 2.5rem; }

.metric-card {

    background: var(--card-bg); padding: 2rem; border-radius: 24px; text-align: center;

    border: 1px solid var(--card-border); position: relative; overflow: hidden;

    box-shadow: var(--shadow-soft); transition: transform 0.3s ease, box-shadow 0.3s ease;

}

.metric-card:hover { transform: translateY(-5px); box-shadow: var(--shadow-hover); }

.metric-value { font-size: 2.5rem; font-weight: 800; margin-bottom: 0.2rem; color: var(--text-
main); }

/* 🐱 Enhanced Hero Section (Card & Illustration) */

.hero {

```

```

display: flex; flex-direction: row; justify-content: space-between; align-items: center;

background: var(--card-bg); padding: 4rem; border-radius: 32px;

box-shadow: var(--shadow-soft); text-align: left; gap: 2rem;

position: relative; overflow: hidden;
}

.hero::after {

  content: ""; position: absolute; top: -100px; right: -50px; width: 400px; height: 400px;

  background: radial-gradient(circle, var(--accent) 0%, transparent 70%); opacity: 0.3; z-index: 0;
  pointer-events: none;
}

.hero-content { flex: 1; position: relative; z-index: 1; }

.hero-image { flex: 1; display: flex; justify-content: center; position: relative; z-index: 1;
animation: float 4s ease-in-out infinite; }

.hero .badge {

  display: inline-flex; align-items: center; gap: 8px; padding: 0.6rem 1.2rem;

  background: var(--input-bg); color: var(--primary); border: 1px solid var(--card-border);

  border-radius: 50px; font-weight: 700; font-size: 0.8rem; margin-bottom: 1.5rem;

  letter-spacing: 1px; text-transform: uppercase;
}

.hero h1 { font-size: 3.5rem; line-height: 1.1; margin-bottom: 0.5rem; color: var(--text-main); }

.hero h2 { font-size: 1.5rem; color: var(--text-muted); margin-bottom: 1.5rem; font-weight: 500;
}

.hero p { font-size: 1.15rem; margin-bottom: 2.5rem; max-width: 550px; color: var(--text-
muted); }

.hero-buttons { display: flex; gap: 1rem; }

/* Custom Scrollbar */

::-webkit-scrollbar { width: 10px; height: 10px; }

::-webkit-scrollbar-track { background: var(--bg-color); }

::-webkit-scrollbar-thumb { background: #C4DDDD; border-radius: 10px; }

::-webkit-scrollbar-thumb:hover { background: var(--primary); }

```

```

/* =====
🧐 SIMULATION RESULTS TARGETED OVERRIDES (MATCHING PIC 1)
===== */

#results th {
  background: #FDE48C; /* Pastel Yellow Header */
  color: #374151;
  border-radius: 0;
}

#results th:first-child { border-radius: 12px 0 0 12px; }
#results th:last-child { border-radius: 0 12px 12px 0; }


#results tbody tr:nth-child(even) td { background-color: #FFFDF0; }
#results tbody tr:nth-child(odd) td { background-color: #FFFFFF; }
#results tbody tr:hover td { filter: brightness(0.97); }


#results .metric-card::before {
  background: linear-gradient(90deg, #FDBA74, #F59E0B); /* Orange Top Line */
}

#results h3 i {
  color: #F5A651 !important; /* Orange Icons */
}


/* Gantt Chart Orange Line & Labels */

.gantt-time-labels {
  display: flex; margin-top: 1.5rem; font-size: 0.85rem; font-weight: 600; color: #4B5563;
  position: relative; min-height: 28px; align-items: center;
}

.gantt-time-labels::before {
  content: ""; position: absolute; left: 0; right: 0; top: 50%; transform: translateY(-50%); height:
  6px; background: #FCA55D; border-radius: 4px; z-index: 0;
}

```



```
.time-label {  
    position: absolute; transform: translateX(-50%); background: #FFFFFF; padding: 2px 10px;  
    border-radius: 8px; border: 1px solid #E5E7EB; z-index: 1; box-shadow: 0 2px 4px rgba(0,0,0,0.02);  
    color: #4B5563;  
}
```

```
/* Responsive */
```

```
@media (max-width: 900px) {  
    .hero { flex-direction: column; text-align: center; padding: 3rem 2rem; }  
    .hero-content { display: flex; flex-direction: column; align-items: center; }  
    .hero h2 { justify-content: center; }  
    .hero-image { margin-top: 2rem; width: 80%; }  
}
```

```
@media (max-width: 768px) {  
    .hero h1 { font-size: 2.5rem; }  
    .hero-buttons { flex-direction: column; width: 100%; max-width: 300px; }  
    .nav-links { flex-wrap: wrap; justify-content: center; gap: 0.25rem; }  
    .nav-links button { padding: 0.4rem 0.8rem; font-size: 0.85rem; }  
    nav { flex-direction: column; gap: 1rem; padding: 1rem; }  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<nav>
```

```
<div class="logo">
```

```
<i class="fas fa-microchip"></i>
```

```
CPU Scheduling Simulator
```

```
</div>
```

```
<div class="nav-links">
```

```
<button class="active" onclick="showPage('home')">Home</button>
```

```
<button onclick="showPage('theory')">Theory</button>
```

```

        <button onclick="showPage('simulator')">Simulator</button>

        <button onclick="showPage('conclusion')">Conclusion</button>

    </div>

</nav>

<section id="home" class="page active">

    <div class="hero">

        <div class="hero-content">

            <div class="badge">OPERATING SYSTEM PRACTICAL</div>

            <h1>CPU Scheduling Simulator</h1>

            <h2 style="color: var(--text-muted);">Shortest Job First (Preemptive)</h2>

            <p>An interactive web-based tool to visualize CPU scheduling and understand process execution step-by-step.</p>

            <div class="hero-buttons">

                <button class="btn" onclick="showPage('simulator')">Run Simulation</button>

                <button class="btn btn-outline" onclick="showPage('theory')">View Theory</button>

            </div>

        </div>

        <div class="hero-image">

            <svg viewBox="0 0 400 350" width="100%" height="auto"
xmlns="http://www.w3.org/2000/svg">

                <path d="M 350 180 C 350 280 280 340 200 340 C 120 340 30 280 40 180 C 50 80 140 40 220 40 C 300 40 350 80 350 180 Z" fill="#FDF4A9" opacity="0.6"/>

                <rect x="80" y="270" width="240" height="12" rx="6" fill="#41ABAE"/>

                <rect x="100" y="282" width="12" height="60" rx="4" fill="#527A6C"/>

                <rect x="288" y="282" width="12" height="60" rx="4" fill="#527A6C"/>

                <polygon points="170,270 240,270 220,220 160,220" fill="#1B4332"/>

                <rect x="140" y="265" width="120" height="5" rx="2" fill="#527A6C"/>

                <path d="M 175 240 L 205 240 M 175 250 L 215 250 M 175 260 L 195 260"
stroke="#38A9AC" stroke-width="3" stroke-linecap="round"/>

                <rect x="310" y="240" width="10" height="100" rx="4" fill="#A4C2C2"/>

```

```

<rect x="290" y="335" width="50" height="8" rx="4" fill="#527A6C"/>
<path d="M 330 230 C 330 260 300 280 270 280 L 240 280 L 250 250 C 270 240 290 200
310 170 C 320 180 330 200 330 230 Z" fill="#38A9AC"/>
<path d="M 270 270 L 240 320 L 260 330" stroke="#E5C29F" stroke-width="14" stroke-
linecap="round" stroke-linejoin="round" fill="none"/>
<path d="M 310 180 Q 280 220 230 240" stroke="#E5C29F" stroke-width="12" stroke-
linecap="round" fill="none"/>
<circle cx="315" cy="130" r="28" fill="#E5C29F"/>
<path d="M 290 130 C 290 90 350 90 350 140 C 350 170 340 180 330 180 C 320 180 330
150 315 150 C 300 150 290 140 290 130 Z" fill="#1B4332"/>
<circle cx="340" cy="160" r="16" fill="#1B4332"/>
<circle cx="60" cy="100" r="8" fill="#38A9AC" opacity="0.5"/>
<circle cx="360" cy="60" r="12" fill="#41ABAE" opacity="0.3"/>
<rect x="80" y="60" width="16" height="16" rx="4" fill="#FDF4A9" transform="rotate(25
80 60)"/>
<rect x="120" y="80" width="80" height="60" rx="8" fill="FFFFFF" stroke="#E8F0F0"
stroke-width="4"/>
<rect x="135" y="110" width="10" height="20" rx="3" fill="#FDF4A9"/>
<rect x="155" y="100" width="10" height="30" rx="3" fill="#41ABAE"/>
<rect x="175" y="90" width="10" height="40" rx="3" fill="#38A9AC"/>
</svg>
</div>
</div>
</section>

```

```

<section id="theory" class="page">
  <h2><i class="fas fa-book-open" style="color: var(--primary);"></i> Theory: Preemptive
  SJF</h2>
  <div class="grid-2">
    <div class="card">
      <h3><i class="fas fa-layer-group" style="color: var(--primary);"></i> The Core Concept</h3>
      <p>Shortest Job First (SJF) is a scheduling policy that selects the waiting process with the
      smallest execution time to execute next. It guarantees the minimum average waiting time among all
      scheduling algorithms.</p>

```

</div>

<div class="card">

<h3><i class="fas fa-exchange-alt" style="color: var(--primary);"></i> Preemptive Mechanics</h3>

<p>Preemptive (SRTF): If a new process arrives with a shorter burst time than the remaining time of the current process, the CPU is taken away (preempted) and given to the new process.</p>

<p>Non-Preemptive: Once a process starts executing, it cannot be interrupted until it finishes its burst time.</p>

</div>

</div>

</section>

<section id="simulator" class="page">

<h2><i class="fas fa-sliders-h" style="color: var(--primary);"></i> Simulator Setup</h2>

<div class="grid-2">

<div class="card">

<h3><i class="fas fa-plus-circle" style="color: var(--primary);"></i> Add New Process</h3>

<form id="processForm" onsubmit="addProcess(event)">

<div class="input-group">

<label>Process Identifier</label>

<input type="text" id="pid" value="P1" required>

</div>

<div class="grid-2" style="gap: 1rem;">

<div class="input-group">

<label>Arrival Time (AT)</label>

<input type="number" id="at" min="0" value="0" required>

</div>

<div class="input-group">

<label>Burst Time (BT)</label>

<input type="number" id="bt" min="1" value="5" required>

</div>

```

    </div>

    <button type="submit" class="btn" style="width: 100%; margin-top: 1rem;">
        Add to Queue
    </button>

</form>
</div>

<div class="card">
    <h3><i class="fas fa-list-ul" style="color: var(--primary);"></i> Ready Queue</h3>
    <table id="inputTable">
        <thead>
            <tr>
                <th>PID</th>
                <th>Arrival</th>
                <th>Burst</th>
                <th>Action</th>
            </tr>
        </thead>
        <tbody>
            <tr><td colspan="4" style="text-align: center; color: var(--text-muted); padding:
2rem;">No processes added yet.</td></tr>
        </tbody>
    </table>
    <br>
    <button class="btn" onclick="runSimulation()" style="width: 100%; background: var(--text-
main); color: white;">
        <i class="fas fa-play" style="font-size: 0.85rem;"></i> Run Simulation
    </button>
</div>
</div>
</section>

```

```
<section id="results" class="page">

  <h2><i class="fas fa-chart-pie" style="color: var(--primary);"></i> Simulation Results</h2>

  <div class="metrics" id="metricsContainer">

    </div>

  <div class="card">

    <h3><i class="fas fa-stream"></i> Execution Timeline (Gantt Chart)</h3>

    <div class="gantt-container">

      <div class="gantt-chart" id="ganttChart"></div>

      <div class="gantt-time-labels" id="ganttLabels"></div>

    </div>

  </div>

  <div class="card">

    <h3><i class="fas fa-table"></i> Detailed Process Metrics</h3>

    <div style="overflow-x: auto;">

      <table>

        <thead>

          <tr>

            <th>Process ID</th>

            <th>Arrival Time</th>

            <th>Burst Time</th>

            <th>Completion (CT)</th>

            <th>Turnaround (TAT)</th>

            <th>Waiting (WT)</th>

          </tr>

        </thead>

        <tbody id="resultTableBody">

          </tbody>

        </table>

      </div>

    </div>

  </div>

</section>
```

```

        </div>
    </div>
    <div style="text-align: center; margin-top: 2rem;">
        <button class="btn btn-outline" onclick="showPage('simulator')">Modify Processes</button>
    </div>
</section>

```

```

<section id="conclusion" class="page">
    <h2><i class="fas fa-clipboard-check" style="color: var(--primary);"></i> Analysis &
    Conclusion</h2>
    <div class="grid-2">
        <div class="card">
            <h3><i class="fas fa-check-circle" style="color: var(--primary); margin-right: 5px;"></i>
            Advantages</h3>
            <ul style="list-style-type: none; color: var(--text-muted);">
                <li style="margin-bottom: 1rem;"><i class="fas fa-circle" style="font-size: 0.5rem; color:
                var(--primary); margin-right: 10px; vertical-align: middle;"></i> Produces the optimal (minimum)
                average waiting time.</li>
                <li style="margin-bottom: 1rem;"><i class="fas fa-circle" style="font-size: 0.5rem; color:
                var(--primary); margin-right: 10px; vertical-align: middle;"></i> Excellent for systems where
                throughput is a priority.</li>
                <li><i class="fas fa-circle" style="font-size: 0.5rem; color: var(--primary); margin-right:
                10px; vertical-align: middle;"></i> Shorter jobs are executed quickly, freeing up memory.</li>
            </ul>
        </div>
        <div class="card">
            <h3><i class="fas fa-exclamation-circle" style="color: var(--text-muted); margin-right:
            5px;"></i> Limitations</h3>
            <ul style="list-style-type: none; color: var(--text-muted);">
                <li style="margin-bottom: 1rem;"><i class="fas fa-circle" style="font-size: 0.5rem; color:
                var(--text-muted); margin-right: 10px; vertical-align: middle;"></i> <strong>Starvation:</strong>
                Long processes may wait indefinitely if short ones keep arriving.</li>
                <li style="margin-bottom: 1rem;"><i class="fas fa-circle" style="font-size: 0.5rem; color:
                var(--text-muted); margin-right: 10px; vertical-align: middle;"></i> Requires precise prediction of
                burst times (difficult in reality).</li>
            </ul>
        </div>
    </div>

```

```
        <li><i class="fas fa-circle" style="font-size: 0.5rem; color: var(--text-muted); margin-right: 10px; vertical-align: middle;"></i> High context switching overhead due to frequent preemptions.</li>
```

```
    </ul>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<script>
```

```
    // --- Navigation Logic ---
```

```
    function showPage(pageId) {
```

```
        document.querySelectorAll('.page').forEach(p => p.classList.remove('active'));
```

```
        document.querySelectorAll('.nav-links button').forEach(b => b.classList.remove('active'));
```

```
        const page = document.getElementById(pageId);
```

```
        page.style.animation = 'none';
```

```
        page.offsetHeight; /* trigger reflow */
```

```
        page.style.animation = null;
```

```
        page.classList.add('active');
```

```
        const buttons = document.querySelectorAll('.nav-links button');
```

```
        buttons.forEach(btn => {
```

```
            if(btn.textContent.toLowerCase().includes(pageId) ||
```

```
                (pageId === 'results' && btn.textContent.toLowerCase() === 'simulator')) {
```

```
            }
```

```
        });
```

```
        event.currentTarget.classList.add('active');
```

```
    }
```

```
    // --- Simulator Logic ---
```

```
    let processes = [];
```

```
    let processCount = 1;
```


// 🥰 Soft Pastel Palette mapped EXACTLY to your uploaded picture 1

```
const colors = ['#78E0A9', '#8EBEFB', '#F6A2A2', '#F9E381', '#A1E6D3', '#E0BBE4', '#FCD7A1'];
```

```
function addProcess(e) {
```

```
  e.preventDefault();
```

```
  const pid = document.getElementById('pid').value;
```

```
  const at = parseInt(document.getElementById('at').value);
```

```
  const bt = parseInt(document.getElementById('bt').value);
```

```
  processes.push({ id: pid, at: at, bt: bt, color: colors[processes.length % colors.length] });
```

```
  updateInputTable();
```

```
  processCount++;
```

```
  document.getElementById('pid').value = `P${processCount}`;
```

```
}
```

```
function removeProcess(index) {
```

```
  processes.splice(index, 1);
```

```
  updateInputTable();
```

```
}
```

```
function updateInputTable() {
```

```
  const tbody = document.querySelector('#inputTable tbody');
```

```
  if(processes.length === 0) {
```

```
    tbody.innerHTML = '<tr><td colspan="4" style="text-align: center; color: var(--text-muted); padding: 2rem;">No processes added yet.</td></tr>';
```

```
    return;
```

```
  }
```

```
  tbody.innerHTML = '';
```

```

processes.forEach((p, index) => {
  tbody.innerHTML += `
    <tr>
      <td>
        <div style="display:flex; align-items:center; gap:12px;">
          <span style="display:inline-block; width:12px; height:12px; background:${p.color};
border-radius:4px;"></span>
          <strong>${p.id}</strong>
        </div>
      </td>
      <td>${p.at} ms</td>
      <td>${p.bt} ms</td>
      <td>
        <button onclick="removeProcess(${index})" style="color: #E53E3E;
background:rgba(229, 62, 62, 0.1); width: 32px; height: 32px; border-radius: 8px; border:none;
cursor:pointer; transition: all 0.2s;">
          <i class="fas fa-trash"></i>
        </button>
      </td>
    </tr>
  `;
});
}

```

```

function runSimulation() {

```

```

  if (processes.length === 0) {
    alert("Please add at least one process.");
    return;
  }

```

```

  let p = processes.map(proc => ({ ...proc, rt: proc.bt, isCompleted: false, ct: 0, tat: 0, wt: 0 }));
  let n = p.length;

```

```

let time = 0;

let completed = 0;

let gantt = [];

let prevId = null;

let contextSwitches = 0;

while (completed < n) {
    let idx = -1;

    let minRt = Infinity;

    for (let i = 0; i < n; i++) {
        if (p[i].at <= time && !p[i].isCompleted) {
            if (p[i].rt < minRt) {
                minRt = p[i].rt;

                idx = i;
            }

            if (p[i].rt === minRt) {
                if (idx === -1 || p[i].at < p[idx].at) {
                    minRt = p[i].rt;

                    idx = i;
                }
            }
        }
    }

    if (idx !== -1) {
        if (prevId !== p[idx].id) {
            if (prevId !== null && prevId !== 'Idle') contextSwitches++;

            gantt.push({ id: p[idx].id, color: p[idx].color, start: time, end: time + 1 });
        } else {
            gantt[gantt.length - 1].end = time + 1;
        }
    }
}

```

```

    }

    p[idx].rt--;
    time++;
    prevId = p[idx].id;

    if (p[idx].rt === 0) {
        p[idx].isCompleted = true;
        p[idx].ct = time;
        p[idx].tat = p[idx].ct - p[idx].at;
        p[idx].wt = p[idx].tat - p[idx].bt;
        completed++;
    }
} else {
    if (prevId !== 'Idle') {
        gantt.push({ id: 'Idle', color: 'transparent', start: time, end: time + 1 });
    } else {
        gantt[gantt.length - 1].end = time + 1;
    }
    time++;
    prevId = 'Idle';
}
}

```

```
renderResults(p, gantt, contextSwitches);
```

```
document.querySelectorAll('.page').forEach(pg => pg.classList.remove('active'));
```

```
document.getElementById('results').classList.add('active');
```

```
document.querySelectorAll('.nav-links button').forEach(btn => btn.classList.remove('active'));
```

```
document.querySelectorAll('.nav-links button')[2].classList.add('active');
```

```
}
```

```
function renderResults(results, gantt, switches) {  
  let totalWT = 0, totalTAT = 0;  
  const tbody = document.getElementById('resultTableBody');  
  tbody.innerHTML = '';  
  
  results.forEach(p => {  
    totalWT += p.wt;  
    totalTAT += p.tat;  
    tbody.innerHTML += `  
      <tr>  
        <td>  
          <div style="display:flex; align-items:center; gap:12px;">  
            <span style="display:inline-block; width:12px; height:12px; background:${p.color};  
border-radius:50%;"></span>  
            <strong>${p.id}</strong>  
          </div>  
        </td>  
        <td>${p.at} ms</td>  
        <td>${p.bt} ms</td>  
        <td style="color: var(--text-main);"><strong>${p.ct} ms</strong></td>  
        <td>${p.tat} ms</td>  
        <td>${p.wt} ms</td>  
      </tr>  
    `;  
  });  
  
  const avgWT = (totalWT / results.length).toFixed(2);  
  const avgTAT = (totalTAT / results.length).toFixed(2);  
  document.getElementById('metricsContainer').innerHTML = `
```

```

    <div class="metric-card">

        <div style="color: #6B7280; font-weight: 600; text-transform: uppercase; font-size:
0.75rem; letter-spacing: 1px; margin-bottom: 5px;">Avg Waiting Time</div>

        <div class="metric-value">${avgWT}<span style="font-size: 1rem; color: #6B7280; font-
weight: 500;"> ms</span></div>

    </div>

    <div class="metric-card">

        <div style="color: #6B7280; font-weight: 600; text-transform: uppercase; font-size:
0.75rem; letter-spacing: 1px; margin-bottom: 5px;">Avg Turnaround Time</div>

        <div class="metric-value">${avgTAT}<span style="font-size: 1rem; color: #6B7280; font-
weight: 500;"> ms</span></div>

    </div>

    <div class="metric-card">

        <div style="color: #6B7280; font-weight: 600; text-transform: uppercase; font-size:
0.75rem; letter-spacing: 1px; margin-bottom: 5px;">Context Switches</div>

        <div class="metric-value">${switches}</div>

    </div>
`;

```

```

const chart = document.getElementById('gantChart');
const labels = document.getElementById('gantLabels');

chart.innerHTML = '';
labels.innerHTML = '';

const totalTime = gantt[gantt.length - 1].end;

labels.innerHTML += `<div class="time-label" style="left: 0%;">0</div>`;

gant.forEach((block, i) => {

    const widthPercent = ((block.end - block.start) / totalTime) * 100;

    const div = document.createElement('div');

    div.className = 'gant-block';

```

```
div.style.width = `0%`;

if(block.id === 'Idle') {
  div.style.background = '#F9FAFB';
  div.style.border = '1px dashed var(--text-muted)';
  div.style.color = 'var(--text-muted)';
  div.innerHTML = '<span style="font-size: 0.8rem;">Idle</span>';
} else {
  div.style.backgroundColor = block.color;
  div.innerHTML = block.id;
}

chart.appendChild(div);

setTimeout(() => { div.style.width = `${widthPercent}%`; }, 150 * i);

const endPos = (block.end / totalTime) * 100;
labels.innerHTML += `<div class="time-label" style="left: ${endPos}%;">${block.end}</div>`;
});
}
</script>
</body>
</html>
```